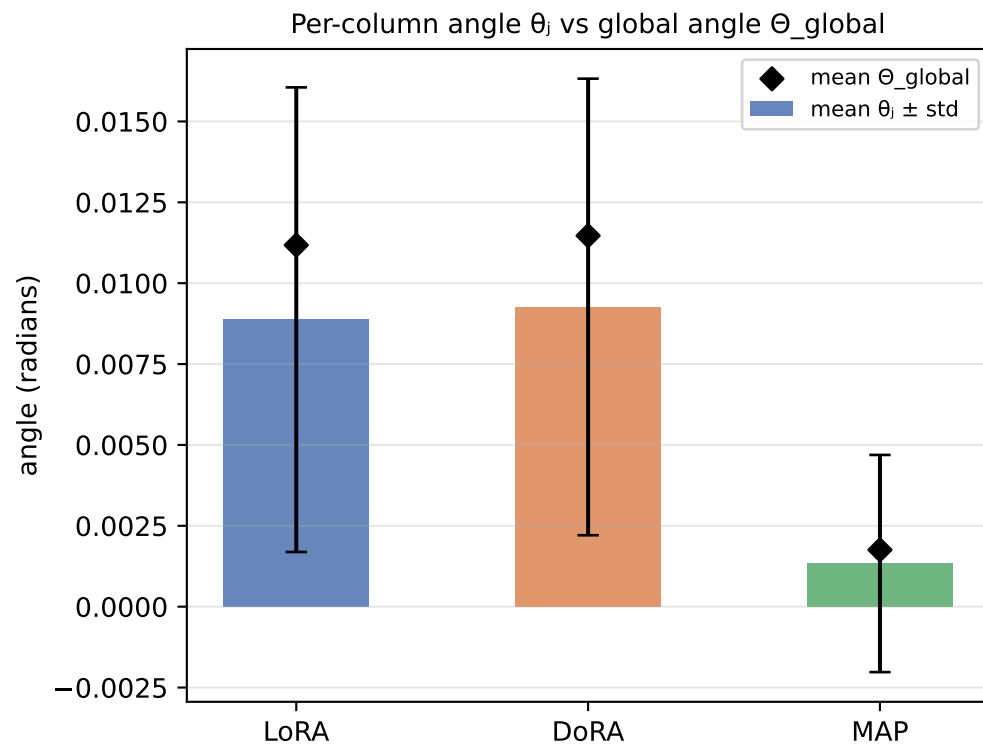
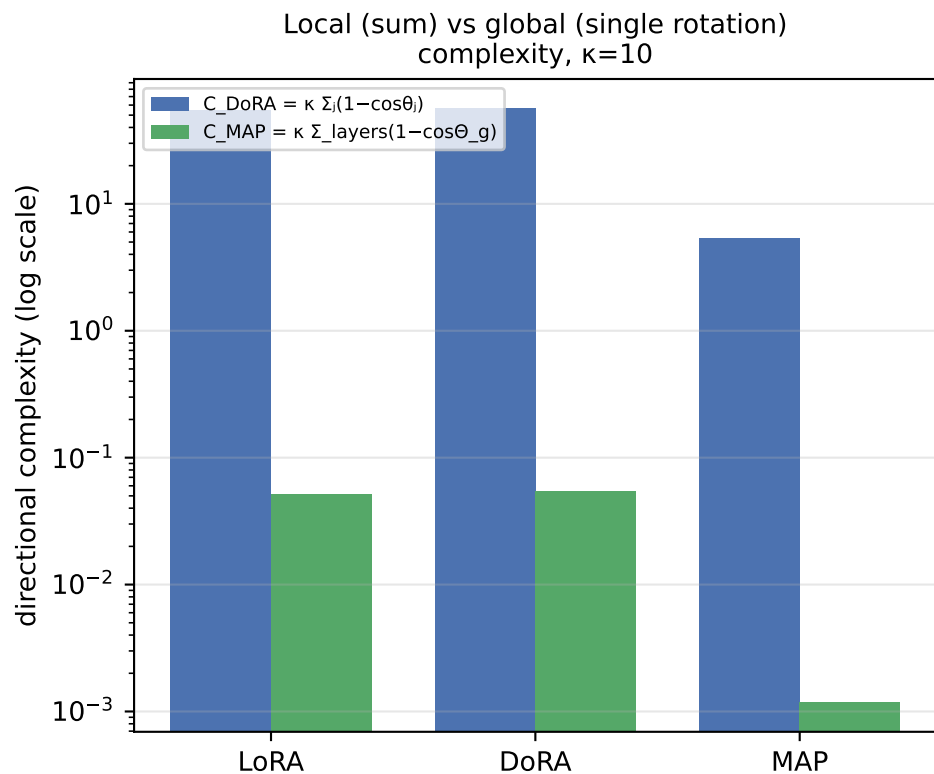


Directional Updates: LoRA vs DoRA vs MAP on RoBERTa-base / SST2

Fine-tuning metrics and aggregate directional complexity

Method	Eval acc	Trainable params	C DoRA (Σ local)	C MAP (global)	mean θ_j (rad)	mean Θ_g (rad)	mean s/d
LoRA	93.12%	1,919,234	54.00	0.052	0.0089	0.0112	0.402
DoRA	92.55%	2,002,178	56.23	0.054	0.0093	0.0115	0.426
MAP	88.53%	1,919,378	5.36	0.001	0.0013	0.0018	0.002



Θ_{global} is the single rotation of the whole flattened matrix (MAP view); θ_j are the per-column rotations (DoRA view).
Exact identity (Thm 3.10): $1 - \cos \Theta_{global} = \text{mean}_j (1 - \cos \theta_j) \rightarrow$ DoRA sums what MAP averages.